



Smart Manufacturing IoT-Cloud Monitoring Dataset – Exploratory Data Analysis (EDA)

Table of contents

Dataset Exploration Report.....	3
Dataset Overview.....	3
Key Features.....	3
Use Cases	3
File Structure.....	4
Purpose of Analysis	4
Data Quality & Structure.....	5
Sensor Behavior and Distributions.....	5
Correlation Analysis	5
Time-Series and Rolling Trends.....	5
Target Analysis: Maintenance vs. No Maintenance.....	6
Failure Type Patterns	6
Key Insights & Takeaways	6

Dataset Exploration Report

Dataset Overview

The **Smart Manufacturing IoT-Cloud Monitoring Dataset** is designed for real-time process monitoring, predictive maintenance, and anomaly detection in industrial environments. It contains simulated IoT sensor readings collected from manufacturing machines, including parameters such as **temperature, vibration, humidity, pressure, and energy consumption**.

This dataset is ideal for developing machine learning models for **predictive maintenance**, reducing downtime, and optimizing operational efficiency. It also includes **anomaly flags, failure types, and a target column (maintenance_required)** indicating whether a machine requires maintenance.

Key Features

- 100,000 sensor records from 50 unique machines
- Timestamps recorded at **1-minute intervals**
- Real-time IoT sensor data: Temperature, Vibration, Humidity, Pressure, Energy Consumption
- Machine operational status: **Idle, Running, or Maintenance**
- Predictive maintenance labels for model training
- Anomaly detection based on extreme sensor readings

Use Cases

-  **Predictive Maintenance** – Forecast failures before they occur
-  **Anomaly Detection** – Identify unusual machine behavior
-  **Smart Manufacturing Optimization** – Enhance production efficiency

File Structure

Column	Type	Description
timestamp	datetime	Minute-by-minute log of each machine reading
machine_id	int	Unique ID for each machine (1–50)
temperature	float	Machine operating temperature (°C)
vibration	float	Vibration intensity level
humidity	float	Environmental humidity (%)
pressure	float	System pressure reading
energy_consumption	float	Energy usage (kWh)
machine_status	int	0 = Idle, 1 = Running, 2 = Maintenance
anomaly_flag	binary	0 = Normal, 1 = Suspicious behavior
predicted_remaining_life	int	Estimated Remaining Useful Life (RUL)
failure_type	category	Failure category (e.g., <i>Normal</i> , <i>Vibration Issue</i> , etc.)
downtime_risk	float	Risk score of potential downtime (0–1)
maintenance_required	binary	Target variable – 1 = Yes, 0 = No

Purpose of Analysis

- Understand sensor behavior and detect patterns in machine performance
- Identify relationships between features and maintenance needs
- Support **predictive maintenance** and **failure prevention** through data-driven insights
- Prepare data for machine learning and anomaly detection models

Data Quality & Structure

- Dataset contains **100,000 clean time-stamped records** across 50 machines.
- **No significant missing values** were found.
- Timestamps were properly parsed and sorted chronologically.
- Additional temporal features (hour, minute, elapsed_time) were engineered to analyze trends and periodic behavior.

Sensor Behavior and Distributions

- Sensor readings such as **temperature, vibration, and energy consumption** display normal or slightly right-skewed distributions.
- **Outliers** were visualized and could be handled using IQR-based clipping (though clipping was not applied in this analysis).
- Overall, readings remain within reasonable operational ranges, reflecting realistic machine behavior.

Correlation Analysis

- **Moderate correlations** exist between temperature, pressure, and energy_consumption, suggesting related physical processes.
- **Vibration** shows weaker correlation with other sensors — indicating it captures unique mechanical conditions.
- No strong multicollinearity detected, meaning all features remain useful for modeling.

Time-Series and Rolling Trends

- Temporal plots reveal **gradual increases** in vibration, pressure, and temperature before maintenance events.
- **Rolling means** smooth noise and highlight long-term drifts, especially in temperature and vibration — clear indicators of potential equipment wear.

Target Analysis: Maintenance vs. No Maintenance

Comparing machines by maintenance_required status reveals strong behavioral differences:

Sensor	Maintenance (Mean ↑)	No Maintenance (Mean ↓)	Insight
Temperature	Higher	Lower	Machines needing maintenance run hotter
Vibration	Higher	Lower	Strong indicator of mechanical wear
Pressure	Slightly higher	Lower	Pressure tends to rise pre-failure
Humidity	Slightly higher	Stable	Environmental influence
Energy Consumption	Higher	Lower	More energy drawn before failure

Conclusion: Machines that require maintenance exhibit consistently **higher sensor readings**, particularly in **temperature, vibration, and energy use** — key predictive features for pre-failure detection.

Failure Type Patterns

- Maintenance-required machines show frequent occurrences of **Vibration Issues** and **Overheating**.
- These patterns confirm that specific failure types are tightly coupled with sensor abnormalities.
- The failure_type column can serve as a strong categorical feature for predictive maintenance models.

Key Insights & Takeaways

- **Core Predictors:** Temperature, Vibration, and Energy Consumption are the most significant early indicators of machine deterioration.
- **Trends Matter:** Rolling averages and time-based analysis help capture slow, progressive damage before failures occur.
- **Data Readiness:** The dataset is clean, structured, and well-prepared for **machine learning applications** such as classification, anomaly detection, or time-series forecasting.

- **Operational Value:** Insights can be used to design real-time maintenance alert systems, optimizing uptime and reducing operational costs.